

PROJECT DESIGN PHASE – SOLUTION REQUIREMENTS

Functional & Non-Functional Requirements

DATE	04/10/2026
TEAM ID	SWTID-2026-8904
PROJECT NAME	AI Study Buddy

MAXIMUM MARKS: 4 Marks

2. Functional Requirements

FR No.	Functional Requirement	Sub Requirement / Story
FR-1	User Registration	Register through form; store user details in MongoDB.
FR-2	User Login	Login with registered email and password; validate credentials.
FR-3	Study Material Management	Create, view, update and delete notes/materials.
FR-4	AI Study Assistance	Enter a question/topic; generate explanations using Google Gemini AI.
FR-5	Quiz Generation	Generate topic-based questions and calculate quiz results.
FR-6	Progress Tracking	Store and retrieve completed topics, quiz scores and study history.
FR-7	User Authentication	Use JWT for secure protected routes.

3. Non-Functional Requirements

NFR	Requirement	Description
NFR-1	Usability	Simple and user-friendly interface for studying, notes, AI chat and quizzes.
NFR-2	Security	Password hashing and JWT-based authentication for protected routes.
NFR-3	Reliability	Store and retrieve study data reliably using MongoDB.
NFR-4	Performance	Provide quick note search and AI response handling.
NFR-5	Availability	Allow access when the server, database and AI service are available.
NFR-6	Scalability	Use modular frontend/backend components for future features.